STA2002: Probability and Statistics II
Contingency Tables: Independence

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Test for

- independence 1 population (only 1 "random experiment" repeated many times)
- homogeneity 2.

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Introduction

- In this lecture we will learn the basics of contingency tables. We first look at a method for testing whether two or more **multinomial distributions** are equal, sometimes called a **test for homogeneity**. Later, we would consider a **test for independence** of attributes of classification. Both of these lead to a similar test statistic. We would also demonstrate the flexibility of the chi-square test learned in class.

- **Suggested reading**: Chapter 9.2 of the textbook.

Introduction

- Now let us suppose that a random experiment results in an outcome that can be classified by **two different attributes**, such as height and weight.
- Assume that the first attribute is assigned to **one and only one** of k **mutually exclusive and exhaustive** event, say $A_1, A_2, \ldots, A_k$.
- While, the second attribute falls into **one and only one** of h **mutually exclusive and exhaustive** events, say, $B_1, B_2, \ldots, B_h$.
- Let the probability of $A_i \cap B_j$ be defined by,

$$p_{ij} := P(A_i \cap B_j), \quad i = 1, 2, \ldots, k, \quad j = 1, 2, \ldots, h.$$

Introduction

- Consider the following contingency table,

A / B	B_1	B_2	$\cdots$	B_h	
A_1	y_{11}	y_{12}	$\cdots$	y_{1h}	$y_{1\cdot}$
A_2	y_{21}	y_{22}	$\cdots$	y_{2h}	$y_{2\cdot}$
$\vdots$	$\cdots$		y_{ij}	$\cdots$	
A_k	y_{k1}	y_{k2}	$\cdots$	y_{kh}	$y_{k\cdot}$
	$y_{\cdot 1}$	$y_{\cdot 2}$		$y_{\cdot h}$	n

- y_{ij} denotes the realized value of Y_{ij}.

Introduction

- The random experiment is to be repeated n independent times, and Y_{ij} denotes the frequency of the event $A_i \cap B_j$, $i = 1, 2, \ldots, k,\ j = 1, 2, \ldots, h$.
- We have that,

$$\sum_{j=1}^{h} \sum_{i=1}^{k} p_{ij} = 1,$$

and

$$\sum_{j=1}^{h} \sum_{i=1}^{k} Y_{ij} = n.$$

- Pay attention: A joint distribution would be proposed for all p_{ij} and Y_{ij}, $i = 1, 2, \ldots, k,\ j = 1, 2, \ldots, h$.

Introduction

- Recall the definition of multinomial distribution.
- Now we assume one multinomial distribution for all p_{ij} and Y_{ij}, $i = 1, 2, \ldots, k,\ j = 1, 2, \ldots, h$.
- Define the following statistic,

$$Q_{kh-1} := \sum_{j=1}^{h} \sum_{i=1}^{k} \frac{(Y_{ij} - np_{ij})^2}{np_{ij}}.$$

- When n is large, Q_{kh-1} approximately follows a chi-square distribution with degrees of freedom $kh - 1$.

Independence

- Suppose that we wish to test the hypothesis of the independence of the A and B attributes, namely,

$$H_0 : \ P(A_i \cap B_j) = P(A_i)P(B_j), \tag{1}$$

for all $i = 1, 2, \ldots, k, \ j = 1, 2, \ldots, h$.

- Question: What is the probability,

$$P(A_i) \quad \text{and} \quad P(B_j),$$

for all $i = 1, 2, \ldots, k, \ j = 1, 2, \ldots, h$?

Independence

- Let us denote $P(A_i)$ by $p_{i\cdot}$ and $P(B_j)$ by $p_{\cdot j}$. Then we have that

$$p_{i\cdot} = \sum_{j=1}^{h} p_{ij} = P(A_i),$$

and

$$p_{\cdot j} = \sum_{i=1}^{k} p_{ij} = P(B_j).$$

- Note that,

$$\sum_{i=1}^{k} p_{i\cdot} = \sum_{j=1}^{h} p_{\cdot j} = 1.$$

Independence

- Now we represent Null Hypothesis H_0 proposed in the expression (1) as follows,

$$H_0 : \ p_{ij} = p_{i\cdot} p_{\cdot j}, \tag{2}$$

for all $i = 1, 2, \ldots, k, \ j = 1, 2, \ldots, h$.

- Question: Think about how to construct a test statistic and the relative natural form of critical region?

Test Statistic

- Recall the statistic,

$$Q_{kh-1} := \sum_{j=1}^{h} \sum_{i=1}^{k} \frac{(Y_{ij} - np_{ij})^2}{np_{ij}}.$$

- Under H_0, $p_{ij} = p_{i\cdot}p_{\cdot j}$, Q is reduced to the following form,

$$Q := \sum_{j=1}^{h} \sum_{i=1}^{k} \frac{(Y_{ij} - np_{i\cdot}p_{\cdot j})^2}{np_{i\cdot}p_{\cdot j}}. \tag{3}$$

- Note that, $p_{i\cdot}$ and $p_{\cdot j}$ are still unknown parameters. Later we use mle to replace these parameters.

Test Statistic

- The mle for $p_{i\cdot}$ and $p_{\cdot j}$ are,

$$\hat{p}_{i\cdot} = \frac{\sum_{j=1}^{h} Y_{ij}}{n} = \frac{Y_{i\cdot}}{n},$$

and

$$\hat{p}_{\cdot j} = \frac{\sum_{i=1}^{k} Y_{ij}}{n} = \frac{Y_{\cdot j}}{n},$$

- Replacing $p_{i\cdot}$ and $p_{\cdot j}$ by their mle in Equation (3) yields,

$$Q := \sum_{j=1}^{h} \sum_{i=1}^{k} \frac{(Y_{ij} - n(Y_{i\cdot}/n)(Y_{\cdot j}/n))^2}{n(Y_{i\cdot}/n)(Y_{\cdot j}/n)}. \tag{4}$$

Test Statistic

- Under H_0, Q has an approximate chi-square distribution with $kh - 1 - (k + h - 2) = (k - 1)(h - 1)$ degrees of freedom.
- The **critical region** with significance level α is,

$$C := \{Q \geq \chi^2_\alpha((k-1)(h-1))\}.$$

- **p-value**,

$$\text{p-value} = P(\chi^2((k-1)(h-1)) > q),$$

where q is the realized value of Q.

Example 9.2-3

Example 1

The 400 undergraduate students in a random sample at the University of Iowa were classified according to the college in which the students were enrolled and according to their gender. The results are recorded in Table 9.2-1, called a $k \times h$ contingency table, where, in this case, $k = 2$ and $h = 5$.

- Question: whether or not the college in which a student enrolls is independent of the gender of that student?

Example 9.2-3

Table 9.2-1 Undergraduates at the University of Iowa

	College					
Gender	Business	Engineering	Liberal Arts	Nursing	Pharmacy	Totals
Male	21	16	145	2	6	190
	(16.625)	(9.5)	(152)	(7.125)	(4.75)	
Female	14	4	175	13	4	210
	(18.375)	(10.5)	(168)	(7.875)	(5.25)	
Totals	35	20	320	15	10	400

Example 9.2-3

- Solution.
  The realized value for the test statistic is

$$q = 18.93.$$

- Under H_0 : Gender and college enrollment are independent, the d.f. of the limiting chi-square distribution of the test statistic Q is

$$(k-1)(h-1) = 1 \times 4 = 4.$$

- If $\alpha = 0.01$, $\chi^2_\alpha(4) = 13.28$. $q > \chi^2_\alpha(4)$. p-value is 0.0008112.
- Decision: Reject H_0. With these data, statistically gender and college are not independent.

Example 9.2-5

Example 2

Two groups of workers have the same qualifications for a particular type of work. Their experience in salaries is summarized by the following 2×5 contingency table, in which the upper bound of each salary range is not included in that listing.

	Salary (Thousands of Dollars)					
Group	27–29	29–31	31–33	33–35	35 and over	Totals
1	6	11	16	14	13	60
2	5	9	8	6	2	30
Totals	11	20	24	20	15	90

Example 9.2-5

- Solution.
  The realized value for the test statistic is

$$q = 4.752.$$

- Under H_0 : The grouping and salary are independent, the d.f. of the limiting chi-square distribution of the test statistic Q is

$$(k-1)(h-1) = 1 \times 4 = 4.$$

- If $\alpha = 0.05$, $\chi^2_\alpha(4) = 9.488$. $q < \chi^2_\alpha(4)$. p-value is 0.314.
- Decision: Retain H_0. That is, with these limited data, group assignment and salaries seem to be independent.